

PREFACE FOR FACULTY

Linear phase is the primary reason engineers choose FIR filters over IIR filters. A filter with strictly linear phase introduces a pure constant time delay across all frequency components, completely preventing dispersion and phase distortion.

Pedagogical Strategy:

1. Formulate the Generalized Linear Phase (GLP) condition: $H(e^{j\omega}) = A(\omega)e^{-j(\alpha\omega - \beta)}$.
 2. Prove that constant group delay requires $\beta = 0$ (symmetry) or $\beta = \pi/2$ (antisymmetry), with group delay $\tau_g = \alpha = \frac{N-1}{2}$.
 3. Systematically categorize the **4 Types of Linear-Phase FIR Filters**:
 - Type I: Symmetric, N odd (Universal).
 - Type II: Symmetric, N even ($H(e^{j\pi}) = 0 \implies$ CANNOT be HPF or BSF).
 - Type III: Antisymmetric, N odd ($H(e^{j0}) = H(e^{j\pi}) = 0 \implies$ BPF, Differentiator, Hilbert).
 - Type IV: Antisymmetric, N even ($H(e^{j0}) = 0 \implies$ HPF, Differentiator, Hilbert).
 4. Prove that roots of linear-phase FIR filters occur in **Conjugate Reciprocal Quadruplets**: $\{z_0, z_0^*, 1/z_0, 1/z_0^*\}$.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** linear phase conditions for symmetric and antisymmetric FIR impulse responses.
 2. **Classify** FIR filters into Types I, II, III, and IV based on length N and symmetry.
 3. **Determine** boundary zero constraints at $\omega = 0$ and $\omega = \pi$ for each FIR type.
 4. **Select** the appropriate FIR type for specific filter applications (LPF, HPF, BPF, BSF, Differentiators, Hilbert Transformers).
 5. **Analyze** zero locations in the complex z -plane.
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2. MATHEMATICAL FOUNDATIONS

2.1 The 4 Types of Linear-Phase FIR Filters

Type	Symmetry	Length N (Order $M = N - 1$)	Amplitude Function $A(\omega)$	Mandatory Boundary Zeros	Suitable Applications
Type I	Symmetric ($h[n] = h[N - 1 - n]$)	N Odd (M Even)	$\sum a[k] \cos(\omega k)$	None	LPF, HPF, BPF, BSF (Universal)
Type II	Symmetric ($h[n] = h[N - 1 - n]$)	N Even (M Odd)	$\sum b[k] \cos(\omega(k - 1/2))$	Zero at $\omega = \pi$ ($z = -1$)	LPF, BPF only
Type III	Antisymmetric ($h[n] = -h[N - 1 - n]$)	N Odd (M Even)	$\sum c[k] \sin(\omega k)$	Zeros at $\omega = 0$ ($z = 1$) & $\omega = \pi$ ($z = -1$)	BPF, Differentiator, Hilbert
Type IV	Antisymmetric ($h[n] = -h[N - 1 - n]$)	N Even (M Odd)	$\sum d[k] \sin(\omega(k - 1/2))$	Zero at $\omega = 0$ ($z = 1$)	HPF, BPF, Differentiator, Hilbert

2.2 Proof of Mandatory Zeros

- **Type II at $\omega = \pi$ ($z = -1$):**

$$H(-1) = \sum_{n=0}^{N-1} h[n](-1)^{-n} = \sum_{n=0}^{N/2-1} h[n][(-1)^{-n} + (-1)^{-(N-1-n)}]$$

Since N is even, $N - 1$ is odd $\implies (-1)^{-(N-1-n)} = -(-1)^{-n} \implies H(-1) = 0$. Therefore, a Type II filter always has a zero at $\omega = \pi$ and can **never** be used as a Highpass or Bandstop filter.

2.3 Zero Quadruplet Locations

Because $h[n]$ is real and symmetric/antisymmetric:

$$H(z) = \pm z^{-(N-1)} H(z^{-1})$$

If $z_0 = re^{j\theta}$ is a zero, then:

1. $z_0^* = re^{-j\theta}$ is a zero (conjugate symmetry for real coefficients).
2. $1/z_0 = \frac{1}{r}e^{-j\theta}$ is a zero (linear phase symmetry).
3. $1/z_0^* = \frac{1}{r}e^{j\theta}$ is a zero. Zeros off the unit circle always occur in groups of 4 (quadruplets).

3. WORKED NUMERICAL EXAMPLES

Example 21.1: FIR Type Identification & Zero Locations

Problem: An FIR filter has transfer function $H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4}$. (a) Determine its FIR Type, filter length N , and group delay τ_g . (b) Can this filter be used as a Highpass filter?

Solution: (a) Coefficients: $h = [1, 2, 3, 2, 1]$.

- Length $N = 5$ (Odd), Order $M = 4$ (Even).

- Symmetry: $h[0] = h[4] = 1, ; h[1] = h[3] = 2, ; h[2] = 3 \implies$ **Symmetric**.
- Therefore, it is a **Type I FIR Filter**.
- Group delay: $\tau_g = \frac{N-1}{2} = \frac{5-1}{2} = 2$ samples.

(b) Highpass Feasibility: Check $H(e^{j\pi}) = H(-1) = 1 - 2 + 3 - 2 + 1 = 1 \neq 0$. Since Type I has no mandatory zeros at $\omega = \pi$, it can theoretically operate as a Highpass filter (though this specific impulse response has $H(e^{j0}) = 9$ and $H(e^{j\pi}) = 1$, making it a Lowpass filter).

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State the four types of linear-phase FIR filters and derive the mandatory boundary zeros for Type II, Type III, and Type IV filters. *(9 Marks)* **(b)** Prove that the roots of a linear-phase FIR filter occur in conjugate reciprocal quadruplets. *(6 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Table of 4 Types with symmetry and length definitions *(3 Marks)*
 - Mathematical proofs showing $H(e^{j\pi}) = 0$ for Type II, $H(1) = H(-1) = 0$ for Type III, and $H(1) = 0$ for Type IV *(6 Marks)*
- **Part (b):**
 - Step-by-step proof using $H(z) = \pm z^{-(N-1)}H(z^{-1})$ and complex conjugation $H^*(z) = H(z^*)$ *(6 Marks)*

5. PYTHON VERIFICATION SCRIPT

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import numpy as np

h = np.array([1, 2, 3, 2, 1])
w = np.linspace(0, np.pi, 500)
H = np.polyval(h[::-1], np.exp(-1j * w))

# Phase should be exactly -2*w
phase = np.unwrap(np.angle(H))
group_delay = -np.diff(phase) / np.diff(w)
print(f"Mean Group Delay: {np.mean(group_delay):.4f} samples (Theoretical = 2.0)")
```